

2.2.2 Bonding and structure

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Ionic bonding	
(a) ionic bonding as electrostatic attraction between positive and negative ions, and the construction of 'dot-and-cross' diagrams	
(b) explanation of the solid structures of giant ionic lattices, resulting from oppositely charged ions strongly attracted in all directions e.g. NaCl	
(c) explanation of the effect of structure and bonding on the physical properties of ionic compounds, including melting and boiling points, solubility and electrical conductivity in solid, liquid and aqueous states	HSW1 Use of ideas about ionic bonding to explain macroscopic properties.
Covalent bonding	
(d) covalent bond as the strong electrostatic attraction between a shared pair of electrons and the nuclei of the bonded atoms	
(e) construction of 'dot-and-cross' diagrams of molecules and ions to describe: (i) single covalent bonding (ii) multiple covalent bonding (iii) dative covalent (coordinate) bonding	'Dot-and-cross' diagrams of up to six electron pairs (including lone pairs) surrounding a central atom.
(f) use of the term <i>average bond enthalpy</i> as a measurement of covalent bond strength	Learners should appreciate that the larger the value of the average bond enthalpy, the stronger the covalent bond. Definition and calculations not required. Average bond enthalpies and related calculations are covered in detail in 3.2.1 f .
The shapes of simple molecules and ions	
(g) the shapes of, and bond angles in, molecules and ions with up to six electron pairs (including lone pairs) surrounding the central atom as predicted by electron pair repulsion, including the relative repulsive strengths of bonded pairs and lone pairs of electrons	M4.1, M4.2 Learners should be able to draw 3-D diagrams to illustrate shapes of molecules and ions. HSW1,2 Using electron pair repulsion theory to predict molecular shapes.
(h) electron pair repulsion to explain the following shapes of molecules and ions: linear, non-linear, trigonal planar, pyramidal, tetrahedral and octahedral	Learners are expected to know that lone pairs repel more than bonded pairs and the bond angles for common examples of each shape including CH ₄ (109.5°), NH ₃ (107°) and H ₂ O (104.5°).

Electronegativity and bond polarity

- (i) electronegativity as the ability of an atom to attract the bonding electrons in a covalent bond; interpretation of Pauling electronegativity values

Learners should be aware that electronegativity increases towards F in the periodic table.

HSW1,2 Using ideas about electronegativity to predict chemical bond type.

- (j) explanation of:

- (i) a polar bond and permanent dipole within molecules containing covalently-bonded atoms with different electronegativities
(ii) a polar molecule and overall dipole in terms of permanent dipole(s) and molecular shape

A polar molecule requires polar bonds with dipoles that do not cancel due to their direction. E.g. H_2O and CO_2 both have polar bonds but only H_2O has an overall dipole.

Intermolecular forces

- (k) intermolecular forces based on permanent dipole–dipole interactions and induced dipole–dipole interactions

Permanent dipole–dipole and induced dipole–dipole interactions can **both** be referred to as van der Waals' forces.

Induced dipole–dipole interactions can also be referred to as London (dispersion) forces.

HSW1,2 Dipole interactions as a model to explain intermolecular bonding.

- (l) hydrogen bonding as intermolecular bonding between molecules containing N, O or F and the H atom of $-\text{NH}$, $-\text{OH}$ or HF

Including the role of lone pairs.

- (m) explanation of anomalous properties of H_2O resulting from hydrogen bonding, e.g.:
(i) the density of ice compared with water
(ii) its relatively high melting and boiling points

HSW1 Use of ideas about hydrogen bonding to explain macroscopic properties.

- (n) explanation of the solid structures of simple molecular lattices, as covalently bonded molecules attracted by intermolecular forces, e.g. I_2 , ice

- (o) explanation of the effect of structure and bonding on the physical properties of covalent compounds with simple molecular lattice structures including melting and boiling points, solubility and electrical conductivity.